LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data
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Objective	Dataset	Model
Predict Outcome: 0 = No diabetes, 1 = Diabetes.	Pima Indians Diabetes Database from Kaggle; file: diabetes.csv.	Median imputation + StandardScaler + balanced Logistic Regression.

Dataset and Workflow

Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Workflow: load Kaggle CSV → inspect data → replace invalid zeros → train-test split (80/20, stratified) → median imputation → scaling → train Logistic Regression → evaluate → interpret coefficients.

Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.734	Overall correct predictions
Precision	0.603	Reliability of positive predictions
Recall	0.704	Actual positive cases detected
F1-score	0.650	Balance of precision and recall
ROC-AUC	0.813	Overall class-separation ability

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

Key Findings

The model reached a ROC-AUC of 0.813, so it separates diabetic and non-diabetic patients fairly well. It correctly found 38 of the 54 diabetic cases in the test set (recall 70.4%), with an overall accuracy of 73.4%. Glucose (1.18) and BMI (0.71) had the strongest positive coefficients, meaning higher values of these raise the predicted chance of diabetes the most. Recall matters in a screening task because the 16 false negatives are people with diabetes who would be missed. The cost is lower precision (0.603), since 25 healthy people were flagged.

Limitation: This small historical dataset (768 rows) represents a specific population. The model is suitable for learning classification concepts, not for medical diagnosis.



